I'll call Alice's spin σ and assign the next letter in the Greek alphabet, τ, to Bob's spin. The full sets of components for Alice's and Bob's spins are

$$\sigma_x,\ \sigma_y,\ \sigma_z$$

and

$$\tau_x,\ \tau_y,\ \tau_z.$$

According to the principles that we laid out earlier, the space of states for the two-spin system is a tensor product. We can make a table of the four states, just as we did in the Interlude. This time, it's a 2 × 2 square, comprising four basis states.

Let's work in a basis in which the *z* components of both spins are specified. The basis vectors are

$$|uu\rangle,\ |ud\rangle,\ |du\rangle,\ |dd\rangle,$$

where the first part of each label represents the state of σ, and the second part represents τ. For example, the first basis vector |*uu*> represents the state in which both spins are up. The vector |*du*> is the state in which Alice's spin is down and Bob's spin is up.

6.5 PRODUCT STATES

The simplest type of state for the composite system is called a *product state*. A product state is the result of completely independent preparations by Alice and Bob, in which each uses his or her own apparatus to prepare a spin. Using explicit notation, suppose Alice prepares her spin in state

$$\alpha_u|u\} + \alpha_d|d\}$$

and Bob prepares his in the state